Classifying 5M bank documents on a Rs 1,00,000 budget

Approach note, IDfy Privy APM take-home · Vansh Gandhi · code, notebook and flowchart in the accompanying repo

The idea

Every document carries a probability for each of the 15 classes. It starts from the class mix expected for its file format (never zero), and each step buys more evidence at a higher price and updates the probabilities with Bayes' rule. A document stops as soon as one class reaches 95%; otherwise it buys the cheapest next piece of evidence that can still separate the classes in play. This is a sequential test (Wald's SPRT in many-class form): money is spent only where the evidence so far cannot decide. A budget controller moves the gates.

The steps, cheapest first

Step	What it reads	Measured cost	Accept rule, and why
L0 Open	bytes, trusted visible text, checksum	3.0 ms/file	Exact duplicate: copy its label. Hidden OCR layers are not trusted (planted text).
L1 Layout	frozen DiT page summary + 6 readable signals (shape, QR, photo...) into a logistic regression	0.11 s/page	≥ 95%: settles 64% of scans at 100.0%. A passbook and a statement share words, not layout.
L2 Text	free text first; for scans a reading ladder: selective read (top and largest lines) → full page → next and last page	0.48 s / 2.2 s a read	≥ 95% after each rung. A word counts only if it is likelier under one remaining class than another, so shared words cancel. Checksum-valid IDs are evidence, not verdicts.
Address question	full read of a KYC card	one full read	Valid PIN + address label or state → PoA, else PoI. PoI vs PoA is a property of the card, not a document type.
L3 LLM	Haiku; text with IDs masked to their type; 2-5 candidate classes + "none of these"; quotes the deciding words	Rs 0.038/call	Its answer is a sensor with a measured error rate; accept at ≥ 95% after the update.
L4 Human	what is still unsure	no price quoted	Last resort; about 2,604 documents at 5M, plus about 600 to check the starting labels.

Critical classes (Legal, Property, PoI, PoA) keep a 95% gate whatever the controller does, once their chance is 10% or more. **Bundled PDFs** are split per page (layout change, first-page look, scanner change), and printed page numbers overrule the layout model: "Page 3" cannot start a document, "Page 1" always does. Every document inside is classified with its page range. Legal and property files open with an e-stamp page that both share, so that bucket reads its later pages before deciding.

Local intelligence

Everything except the LLM runs on CPU: a frozen, pretrained document-vision model (DiT) with a classifier trained in seconds; a text model over 3-5 letter pieces (it survives OCR typos and run-together words) plus checksummed Indian IDs (Aadhaar, PAN, GSTIN, passport MRZ); a calibrated address rule; a page-by-page bundle splitter. Each is temperature-calibrated, so 95% means 95%. Together they settle 99.5% of documents before any LLM call.

How much the LLM should think: measured, not assumed

On 150 hard documents (those L2 sent on or settled below 99%): label-only 91.3% at Rs 0.014 a call; quote-the-evidence 98.7% at Rs 0.038; full thinking the same accuracy at a little more. The rule takes the cheapest mode whose accuracy is at least 95% with 90% confidence: quote-the-evidence. Calls ran through a Claude subscription (*claude -p*, tools off), priced at the brief's token rates.

Results (synthetic documents)

Test set	Macro-F1 (humans as errors / left out)	Critical recall	To LLM / humans
Dev-test (843 documents)	0.998 / 0.999	PoI 1.00, PoA 1.00, Legal 1.00, Property 1.00	0.4% / 0.1%
Lockbox A: sealed, run once; 40% in a layout never seen	0.990 / 0.994	PoI 1.00, PoA 0.98, Legal 0.95, Property 1.00	10 / 6 of 881
Lockbox C: fresh, from the final generator, built after everything was frozen, run once	0.982 / 0.984	PoI 1.00, PoA 1.00, Legal 1.00, Property 1.00	3 / 1 of 430
Lockbox B: 120 statements from another generator (AgamiAI, Hugging Face), layout-only splitter, run once	The splitter cut each 6-page statement into 2.53 pieces; 81% of pieces came out as statements.		
Lockbox D: 120 fresh AgamiAI statements, with page numbers, run once	Every statement kept whole (120/120); 114/120 labelled bank statement; 26 LLM calls, 4 to a person.		

No lockbox was used to train, tune or choose anything; each was run once. Lockboxes A and C pass the bar (macro-F1 ≥ 0.85, critical recall ≥ 0.90); unseen-layout documents scored 0.980 (A) and 0.978 (C). Lockbox B exposed one weakness: the layout-only splitter cut unfamiliar statements into pieces. The page-number fix was built on 80 other fresh statements (kept whole: 16/80 before, 80/80 after; our own bundles unchanged) and checked once on Lockbox D.

Cost at 5M documents (cloud vCPU taken as 2x slower than the laptop core measured)

Item	Cost
L0 + L1 on every page (render, signals, DiT on CPU)	Rs 2,343
Selective read of every scanned page; page-number lines of multi-page scans	Rs 9,809
Further reads (full pages, later pages, KYC address reads)	Rs 8,280
LLM (0.62% of documents, Rs 0.038 a call)	Rs 1,169
Humans (about 2,604 documents; the brief quotes no price)	Rs 0
Total (0.43 paise a document); 95% interval from measurement	**Rs 21,601** (Rs 20,449 to Rs 22,766)
With a scan-heavy stretch (simulated run below)	Rs 24,192
Our target: Rs 50k (a 50% margin); the controller aims at Rs 45k after a 10% reserve	Brute force (OCR all, LLM all): Rs 130,170

How precise. Measured inputs (a fresh single-core timing sample, per-document rates resampled within class, LLM cost from token counts) put the total within 5.4% either way at 95%. Assumptions move it more: a cloud vCPU as fast as the laptop Rs 11,385, 3x slower Rs 31,818; the brief's 350 KB typical PDF fits 1.7 to 6.2 pages (Rs 18,358 to Rs 29,585); the worst combination is Rs 43,794, still under the Rs 50k target.

Before a real run, L0 over the whole bucket (about Rs 225) gives exact page counts, and timing the first 1,000 documents on the real machine gives the cloud speed.

Budget controller. After each batch it compares spend with a straight plan line and moves one gate, τ, which applies to every gate before the LLM (reading is where the money goes). Measured trade-off at 5M: τ 0.85 costs Rs 20,678 at 0.979 accuracy, 0.95 Rs 21,601 at 0.998, 0.99 Rs 24,337 at 0.996. Raising τ bought no accuracy, so under budget the controller holds 95% and the rest stays as margin. Squeezed to Rs 23k, it lowered τ to 0.86 during the scan-heavy stretch and spent Rs 23,753 against Rs 24,192 without it (chart). Most cost is fixed reading, so a much tighter budget needs a person to sign off on lower accuracy.

Assumptions (stated instead of asking)

- 15 classes (the brief also says "25" once). The class mix per format is assumed, matching the brief's format split within 0.03 points; KYC is 20%.
- Critical: Legal, Property, PoI, PoA ("regulatory" read as KYC; no class is medical). PoA if an address is visible, else PoI; a Voter ID front counts as PoI (the brief lists it under PoA only). Passbooks are not statements. Anything that transfers, mortgages, titles or values property is Property; other agreements, rent included, are Legal. An email takes the class of its attachment.
- CPU at the brief's Rs 32/hr for 8 vCPU; LLM tokens at the brief's rates; human review has no quoted price, so it is counted, not costed, and kept in the low thousands. A real run first needs a labelled seed from the client's bucket: the LLM labels about 5,000 documents and people check only where it disagrees with our models (about 600).

Limitations and next steps

- All accuracy is on synthetic documents; ID cards and marksheets follow the official designs but were not checked against real copies. The only outside source tested is bank statements (Lockboxes B and D).
- The masker removes PAN 100%, DL 100%, email 100%, Aadhaar 88%, phone 85%, date of birth 62% of such spans in the Maskara Indian PII test set (Hugging Face); names are not masked yet (next: a name model). The KanoonGPT legal set was not used: it is mostly statutes and notifications, not contracts or deeds.
- Next: a name masker; a small fine-tuned text model (LoRA) for L2; conformal prediction for a recall guarantee on critical classes; a vision LLM for scans whose text cannot be read; more outside document types for the splitter.

The controller at work: budget squeezed to Rs 23k, scan-heavy batches 41-56

